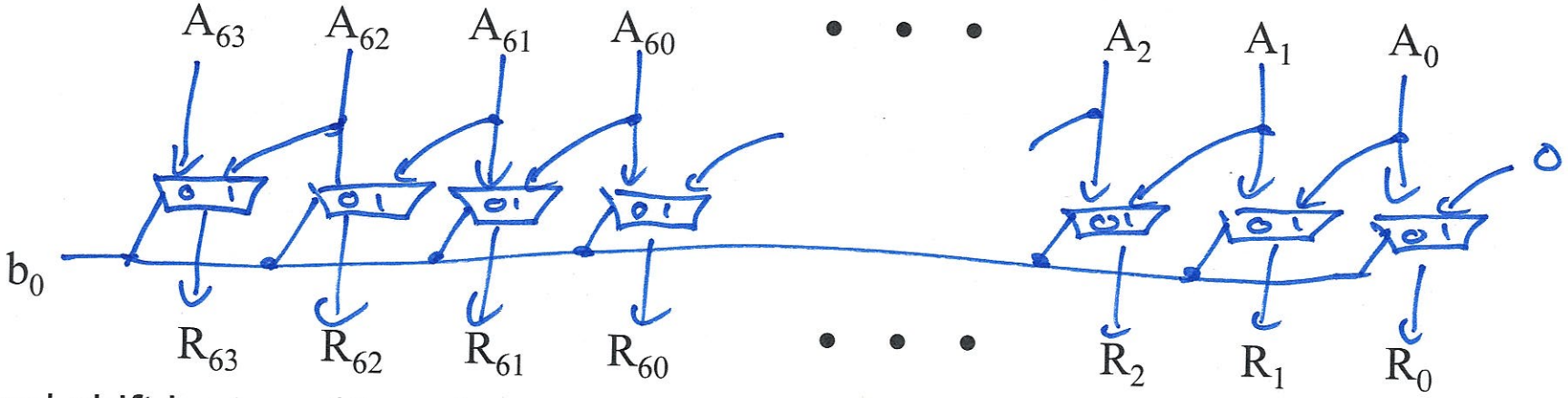
Review Problem 12

- How would the ALU's flags be used to help with each of the following branches? The first is filled in for you:
  - B.EQ: SUBS X31, <val1>, <val2>; use zero flag
  - B.NE: " " " " use $\overline{\text{zero}}$
  - B.GE: " " " " $\overline{\text{negative} \oplus \text{overflow}}$
  - B.GT: " " " " $\overline{(\text{negative} \oplus \text{overflow})} * \overline{\text{zero}}$
  - B.LE: " " " " $(\text{negative} \oplus \text{overflow}) + \text{zero}$
  - B.LT: " " " " $\text{negative} \oplus \text{overflow}$

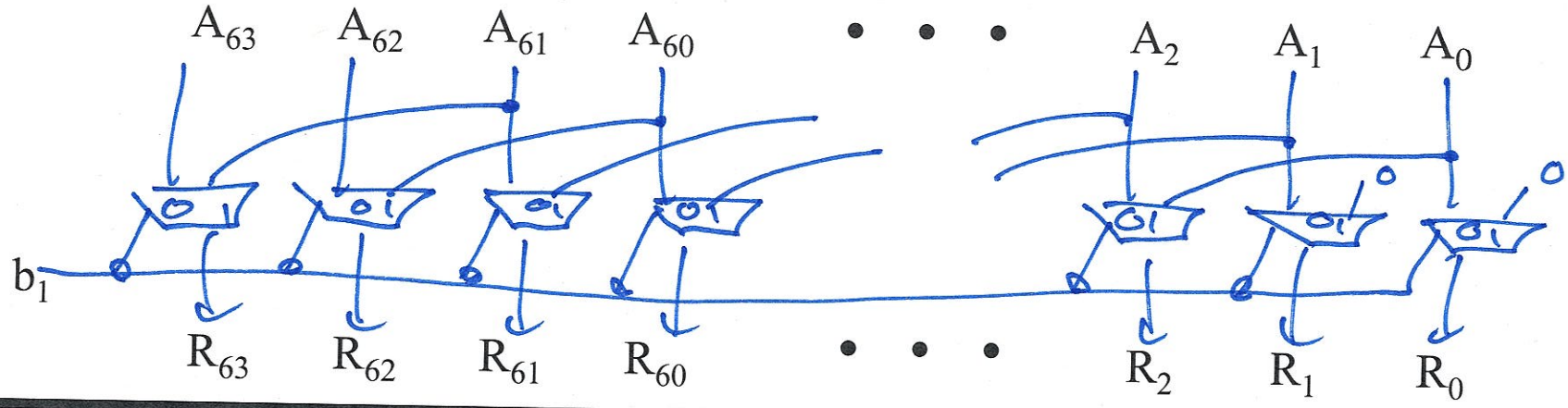
Shifter

Support shift operations: (A << 001101)

Optional shift by one: (A << b_0)

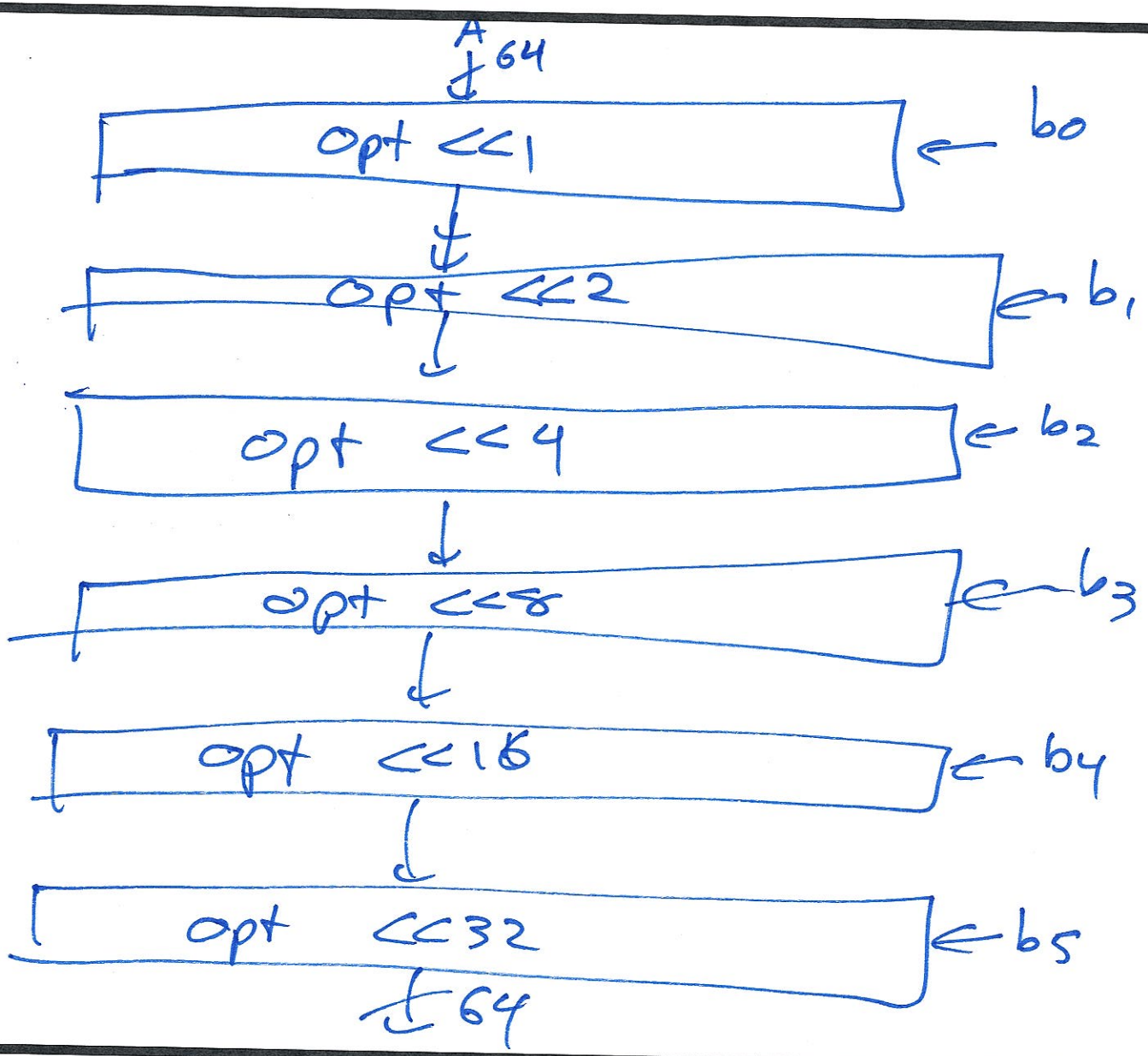


Optional shift by two: (A << b_1)



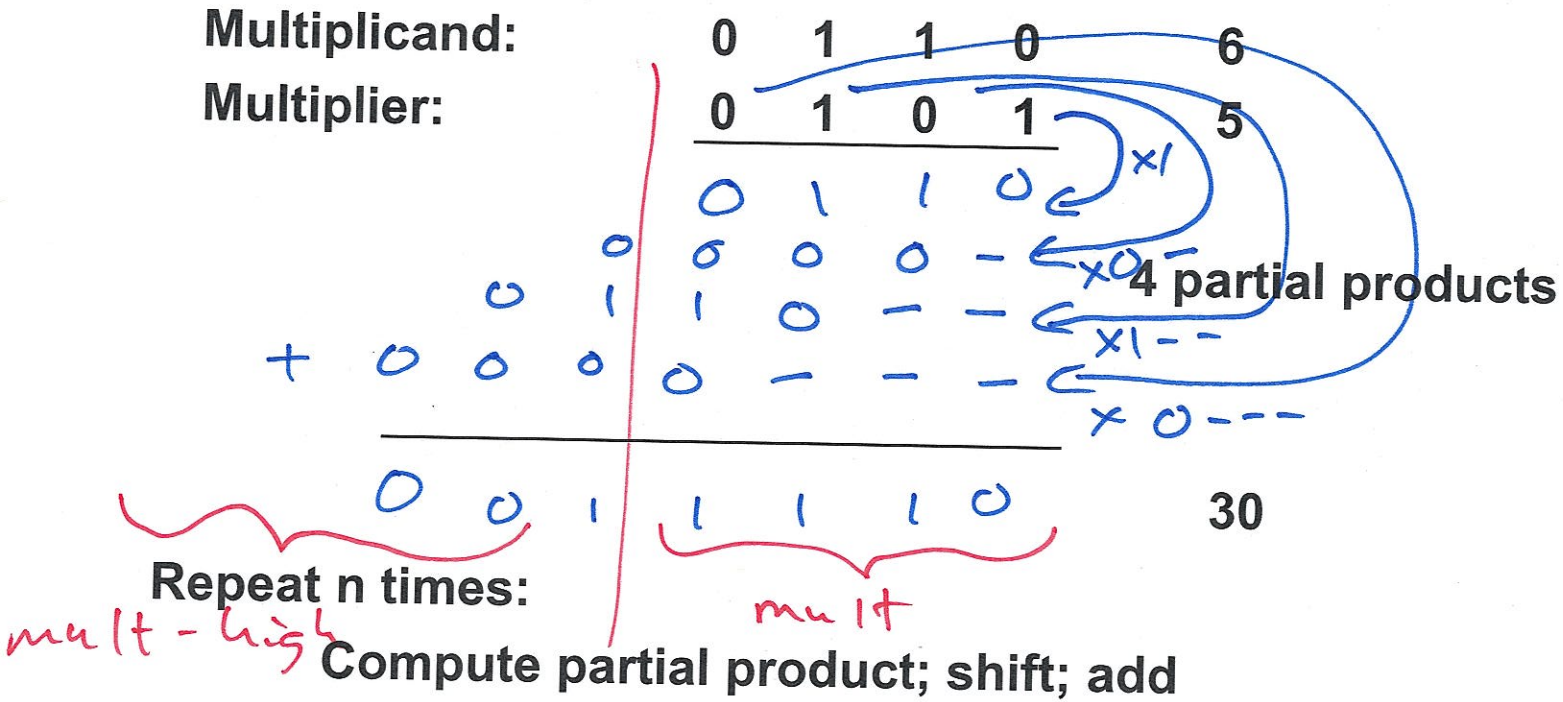
A <- distance

Shifter (cont.)



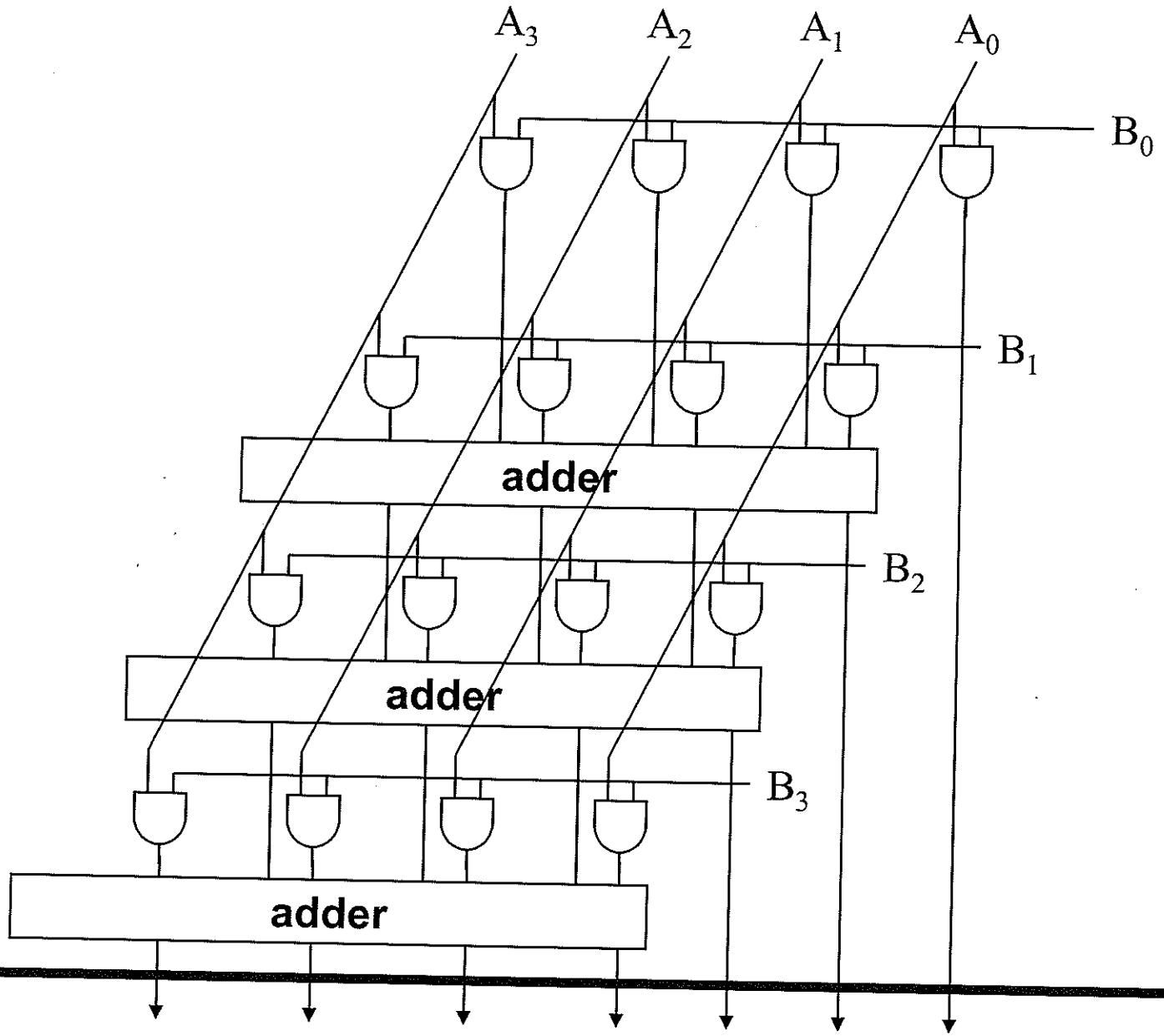
Multiplication

Example



NOTE: Each bit of partial products is just an AND operation

Parallel Multipliers



Computer "Performance"

Readings: 1.6-1.8

BIPS (Billion Instructions Per Second) vs. GHz (Giga Cycles Per Second)

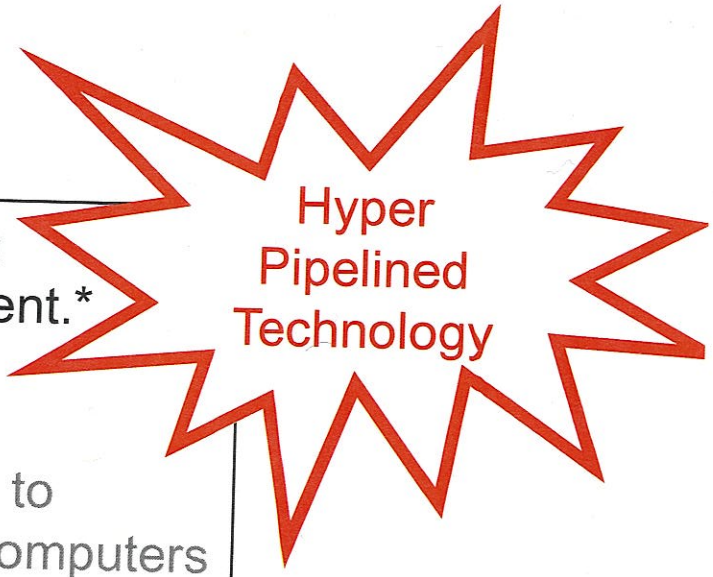
Throughput (jobs/seconds) vs. Latency (time to complete a job)

Measuring "best" in a computer



The PowerBook G4 outguns Pentium III-based notebooks by up to 30 percent.*

* Based on Adobe Photoshop tests comparing a 500MHz PowerBook G4 to 850MHz Pentium III-based portable computers



Performance Example: Homebuilders

Builder	Time per House	Houses Per Month	House Options	Dollars Per House
Self-build	24 months	1/24	Infinite	$200,000
Contractor	3 months	1	100	$400,000
Prefab	6 months	1,000	1	$250,000

Which is the "best" home builder?

Homeowner on a budget? DPH → Self-build

Rebuilding Haiti? HPM → Prefab

Moving to wilds of Alaska? TPH → Contractor

Which is the "speediest" builder?

Latency: how fast is one house built?

Throughput: how long will it take to build a large number of houses?